🥑 AvocadoBook toy laptop – print sheet (Anycubic Kobra X + ACE)

Starting values: Anycubic does not publish a dedicated PETG profile for the Kobra X. Start from the **Kobra X + PETG** profile of Anycubic Slicer Next (version ≥ 1.3.9.3) and change only the items below. Speeds not listed, pressure advance and retraction: keep the profile values. **Everything below is already set in the project file** `AvocadoBook_progetto.3mf`.

1. Materials and ACE slots

Slot	Material	Colour	Used for
1	PETG	Grey	bottom, top case, trackpad, lid body
2	PETG	White	key legends, lid logo, S-springs
3	PETG	Black	keys, screen, screen bezel, text under the bottom
4	PLA	any colour	small hinge support (plate 3) and sheets under the springs (plate 6) only

Project file `AvocadoBook_progetto.3mf`: 7 plates, slots and the settings of this sheet already set. Every 3MF has **one group per material** ("1 GRIGIO PETG" = grey, "2 BIANCO PETG" = white, "3 NERO PETG" = black, "4 PLA (supporti)" = PLA supports): assign the filament to the group, not to single parts. On the printer, the type of each slot must match the slicer (PETG/PLA). Spools without RFID: set type and colour of the slot by hand.

2. Filament settings

PETG grey, white, black	
Nozzle	230 °C, first layer included (it must stay inside the PLA range, because some plates use both)
Bed	80 °C for the whole print (the lid warped at 70 °C)
Part fan	min 10 % – max 30 %, off for the first 5 layers (less cooling = less warping)
Flow ratio	0.95–0.98 (best after a calibration)
Max volumetric speed	12 mm³/s
Z-hop	0.4 mm (fewer scratches and strings)
Drying	65 °C for 4–6 h before long prints
Smooth PEI plate (print aid, not part of the toy)	a thin layer of glue stick as release agent

PLA supports only	
Nozzle	225 °C (inside the PETG range)
Bed	70 °C
Part fan	60–100 %
Interface	PLA and PETG do not bond: the supports come off clean by hand. The small gap is already in the model.

3. Process settings (all plates)

Layer height	0.20 mm (first layer 0.20 mm)
Walls	4 (keys and bezel: 3)
Top / bottom layers	5 / 4
Infill	gyroid 15 % (top case 30 %, lid and trackpad 20 %, springs 100 %)
Wall generator	Arachne (default) – needed for thin ribs and leaves
Elephant foot compensation	0.15 mm
Hole / X-Y contour compensation	0 / 0 (tolerances are built into the model)

Supports	**off** (the only support needed is already a part of the model)
Ironing	**off everywhere**: with PETG it drags blobs over the surface. Plate **2_piano**: 8 top layers, 30 % infill, monotonic top surface 0.4 mm at 40 mm/s.
Brim	**none on any plate** (a PETG brim on textured PEI only comes off with scissors); the large flat parts have tear-off "mouse ears" built in
Speed (quality first)	first layer 25 mm/s (it is the visible face of the parts printed face down); outer wall and top surfaces 60 mm/s; inner walls 120 mm/s; outer wall and top surface acceleration 1000 mm/s²; precise outer wall on
Seam	aligned / rear
Prime tower	only on the multicolour plates (1_fondo, 3_coperchio, 4_tasti, 6_molle_S); width 35 mm. **Position: back left**, e.g. X ≈ 15, Y ≈ 200: on those plates the area X 0–85 / Y 182–260 is kept free on purpose. If the slicer re-centres the objects on import, move them to the front right first.
Flushing volumes	black → white: increase (multiplier 1.2–1.5); grey ↔ black: default; to/from PLA: default

4. Plates

Plate	Object → filament	Notes
1_fondo (bottom) 2 colours	A_fondo → 1 grey A_fondo_scritta_nera → 3 black	The text "AvocadoBook – Designed in Italy" under the bottom is black, in the first 3 layers (face on the bed): prime tower back left, only for those layers. No brim. Ribs with the key-tab channels, spring slots and trackpad click leaves: do not reduce the walls, no supports.
2_piano (top case) 1 colour	B_piano → 1 grey	4 tear-off ears at the corners (snap them off by hand). **The two hinge covers are printed solid**: the project has two modifiers (*MOD_cerniera_piena_1/2*: 100 % infill, 6 walls); do not delete them (the separate file 2_piano.3mf does NOT have them). Above the axle the cover forms a small arch: no supports. No ironing. **Remove it with a cold bed by flexing the plate, never by pulling on the hinge covers.**
3_coperchio (lid) 4 materials	C_coperchio_corpo → 1 grey C_coperchio_schermo → 3 black C_coperchio_logo → 2 white C_supporto_PLA → 4 PLA	Printed with the outer face on the bed: the logo is in the first 4 layers. **6 tear-off ears** (corners and next to the hinge notches, 4 mm neck): bend them back and forth by hand, file the small leftover. **Essential: cover the printer** (an upside-down cardboard box or a tent) and keep it away from drafts: the lid is the part most prone to warping. Remove the PLA support by hand after printing.
4_tasti (keys) 2 colours	D_tasti → 3 black D_lettere → 2 white	Hollow keys printed face down: legends in the first 3 layers, smooth face. 3 walls. Increase black → white flushing, otherwise the legends turn grey. The key tabs stick out below the edge: no supports. No brim. First layer 20 mm/s.
5_cornice_schermo (screen bezel) 1 colour	E_cornice → 3 black	Visible face on the bed: clean bed, slow first layer. 4 tear-off ears. 3 walls. Pegs: minimum layer time ≥ 5 s so they don't deform.
6_molle_S (springs) 2 materials	F_molle_S_… → 2 white F_fogli_PLA → 4 PLA	12 spring strips, each on a **0.4 mm PLA sheet**: PETG does not bond to PLA, so the springs come off by hand, no ears and no brim. 100 % infill, 2 walls, Arachne. Prime tower back left.
7_trackpad 1 colour	B_trackpad → 1 grey	Trackpad face down, **no supports**: the rim and the guide fins have 45° top faces and print by themselves. 4 walls, 20 % infill.

Part names in the files are in Italian: fondo = bottom, piano = top case, coperchio = lid, tasti = keys, lettere = legends, cornice = bezel, molle = springs, fogli = sheets, supporto = support, scritta nera = black text.

5. Assembly (summary – see the assembly instructions)

1. In the bottom, push the 10 spring strips **upright** into their slots, rail down: one front and one back strip per row (the back one is mirrored). The file name gives row and side (fila 1 = number row; davanti = front, dietro = back).
2. Place every key: its 4 tabs slide into the channels of the ribs. Trackpad: insert the 2 short strips, then place the trackpad.
3. Lid: slide the hinge axle from the front into the two cradles of the bottom and **open it to at least 105°** (at 90° the screen gets in the way of the top case).
4. Lower the top case from above: the two hinge covers go over the axle. Press at the back first, then towards the front, until all hooks click.

6. Quick fixes

Problem	What to try
Strings / rough surfaces	dry the PETG; nozzle −5 °C (not below 225 °C on plates with PLA); Z-hop 0.4

Parts that don't fit	elephant foot compensation 0.15; check the printer's dimensional calibration
Greyish legends / logo	increase black → white flushing
Prime tower comes loose	make it wider (more bed contact); clean the bed
Layers splitting (brittle hooks)	lower fan; nozzle +5 °C
PETG tearing the bed surface	glue stick as release agent